

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2456064.8001 +/- 0.0027 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.181 +/- 0.01
Transit depth (Rp/Rs)^2	3.29 +/- 0.36 [%]
Orbital Inclination (inc)	87.0 +/- 0.55
Airmass coefficient 1 (a1)	1.572 +/- 0.017
Airmass coefficient 2 (a2)	-0.294 +/- 0.015
Scatter in the residuals of the lightcurve fit is	1.07 %
Best Comparison Star	None
Optimal Aperture	4.63
Optimal Annulus	8.25
Transit Duration (day)	0.1145 +/- 0.0077

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.181 $\pm$ 0.01 vs 0.1457 (deviation 3.53 σ)
Duration fit vs published	0.1145 d vs 0.12 d (deviation 0.71 σ)
Mid-time O-C	-13.5 min
Depth SNR	18.1
Residual scatter	1.07 %

Light curve

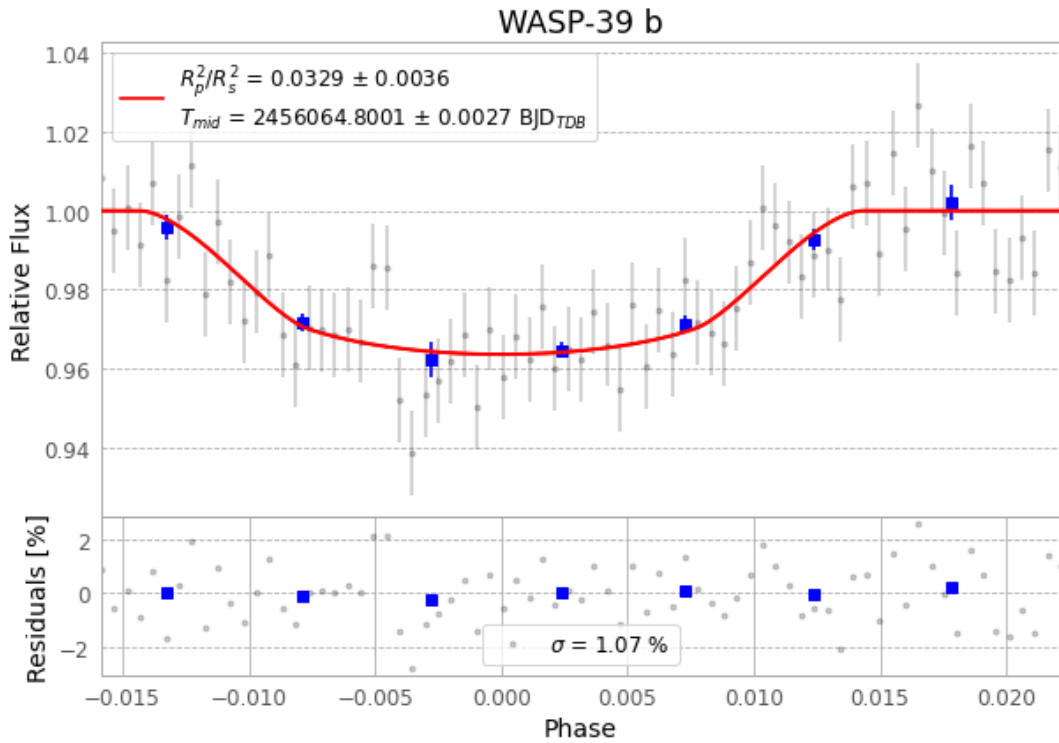


Figure 1 — FinalLightCurve_WASP-39 b_2012-05-16.png (SHA-256 07e5f7f97c230f07...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [313, 280], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 08cb693118ce2c64...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

75 FITS frames (49 MB), WASP-39120517053013.FITS → WASP-39120517091212.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-39-120517.log (64be0d2bea07...), FinalLightCurve_WASP-39 b_2012-05-16.png (07e5f7f97c23...), FinalLightCurve_WASP-39 b_2012-05-16.csv (90e3de76ecc4...)

Compute resources

Wall time 213.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-09 00:58Z.